CINEMA MANAGEMENT SYSTEM

TA: RAMADAN WASFI

TEAM #: 91

SC – Level 3	2022170462	ندى ايمن محمد اسماعيل
SC – Level 3	2022170145	رانا ناصر محمد جودة
SC – Level 3	2022170453	مي عز الدين سعد حنفي
SC – Level 3	2022170323	لوسيا عونى جرجس حنالله
SC – Level 3	2022170472	نوران احمد سمير احمد
SC – Level 3	2022170189	سلمي عماد احمد علي

Contents

Introduction

The **Cinema Management System (CMS)** is designed to streamline and automate the core operations of a cinema, enhancing efficiency and customer experience. The system aims to facilitate **ticket booking, movie scheduling, customer management, and sales tracking**, reducing manual workload and improving overall operational effectiveness.

The CMS will serve as a centralized platform including various functionalities essential for cinema operations. Key features include:

- **Movie Scheduling**: Cinema staff require an intuitive interface to create, update, and upload movie schedules.
- **Ticket Booking:** Online ticket purchasing capabilities with real-time seat selection.
- **Customer Management:** A database to store customer information and track preferences.

The CMS will integrate seamlessly with:

- **Payment Gateways**: For secure online transactions.
- **Marketing Platforms:** Enabling targeted promotions and communication with customers.

User Requirements

Cinema Staff

- Ability to **add, modify, and manage** movie schedules.
- **Set movies genres**, durations, age restrictions and descriptions.
- **Assign movies** to specific screens and time slots.
- Management of **ticket bookings** and customer inquiries at the box office.
- **Real-time access** of bookings made.
- **Manage** different **shows** and screens.
- **Approving a refund payment** to cancel a user booking.

Customer

- Easy-to-use **online ticket booking system** with secure payments.
- Ability to **view show times**, select seats, and reserve tickets in advance.
- Select **preferred available seats**.
- Ability to **cancel seat booking** and refund it money.
- **Search** for a preferred movie and **see its details**.

Functional Requirements

Select Movie:

1. The system should allow the user to view detailed information about a chosen movie.
2. The system requires the user to select a movie from the available list.
3. The movie selection input comes from the user's interaction with the movie listings.
4. The system must have at least one movie available in the database.
5. After execution, the system displays the complete movie details to the user.
6. The system outputs the movie's title, description, duration, ratings, and showtimes.

Search Movies:

1. The system should enable users to search for movies based on various criteria.
2. The system requires the user to enter a search query (title, genre, etc.).
3. The search input is provided by the user via the search interface.
4. The movie database must contain at least one movie record.
5. After execution, the system filters and retrieves matching movies.
6. The system outputs a list of movies that match the search criteria.

Make a Reservation:

1. The system should allow customers to reserve seats for a selected movie showtime.
2. The system requires the movie selection, showtime, seat choice, and user credentials.
3. Inputs come from the user's selections and the seat availability database.
4. The user must be logged in, and seats must be available for the chosen showtime.
5. After execution, the system reserves the seats and updates the booking database.
6. The system outputs a booking confirmation with ticket details.

Manage Show:

1. The system should allow admins to schedule or modify movie showtimes.
2. The system requires the movie ID, screen number, date, and time inputs.
3. Inputs are provided by the admin via the scheduling interface.
4. The admin must be logged in with appropriate privileges, and the movie must exist.
5. After execution, the system updates the showtime database.
6. The system outputs the updated schedule for the affected movie.

Manage Movies:

1. The system should allow admins to update or modify movie details.
2. The system requires the movie ID and updated metadata (description, trailer, etc.).
3. Inputs are provided by the admin via the movie management interface.
4. The admin must be logged in with sufficient permissions, and the movie must exist.
5. After execution, the system updates the movie database with new details.
6. The system outputs the updated movie details.

Non-Functional Requirements

Reliability (75%):
The system should be **highly dependable** to ensure **smooth ticket bookings** and cinema operations. It should have minimal downtime and be resilient to failures. This includes features like redundancy, fault tolerance, and robust error-handling mechanisms

Security (90%):
Since the system **handles sensitive data** such as payment information and customer details, it must be protected against cyber threats and fraud. Strong encryption protocols, strict access controls, and regular security audits should be implemented to safeguard data.

Performance (85%):
The system should be **highly responsive**, ensuring that page loading and critical operations (such as booking and payment) **take no longer than two seconds**. This involves optimizing database queries, reducing network latency, and conducting regular load testing

Availability (80%):
The system should be **available 24/7 without interruptions** that affect ticket bookings or cinema management. This can be achieved by implementing redundant components, failover mechanisms, and disaster recovery plans

Usability (75%):
The system should have an **intuitive user interface** for both customers and staff, making ticket booking and schedule management easy. The interface design should prioritize simplicity, clarity, and **ease of navigation for a smooth user experience.**

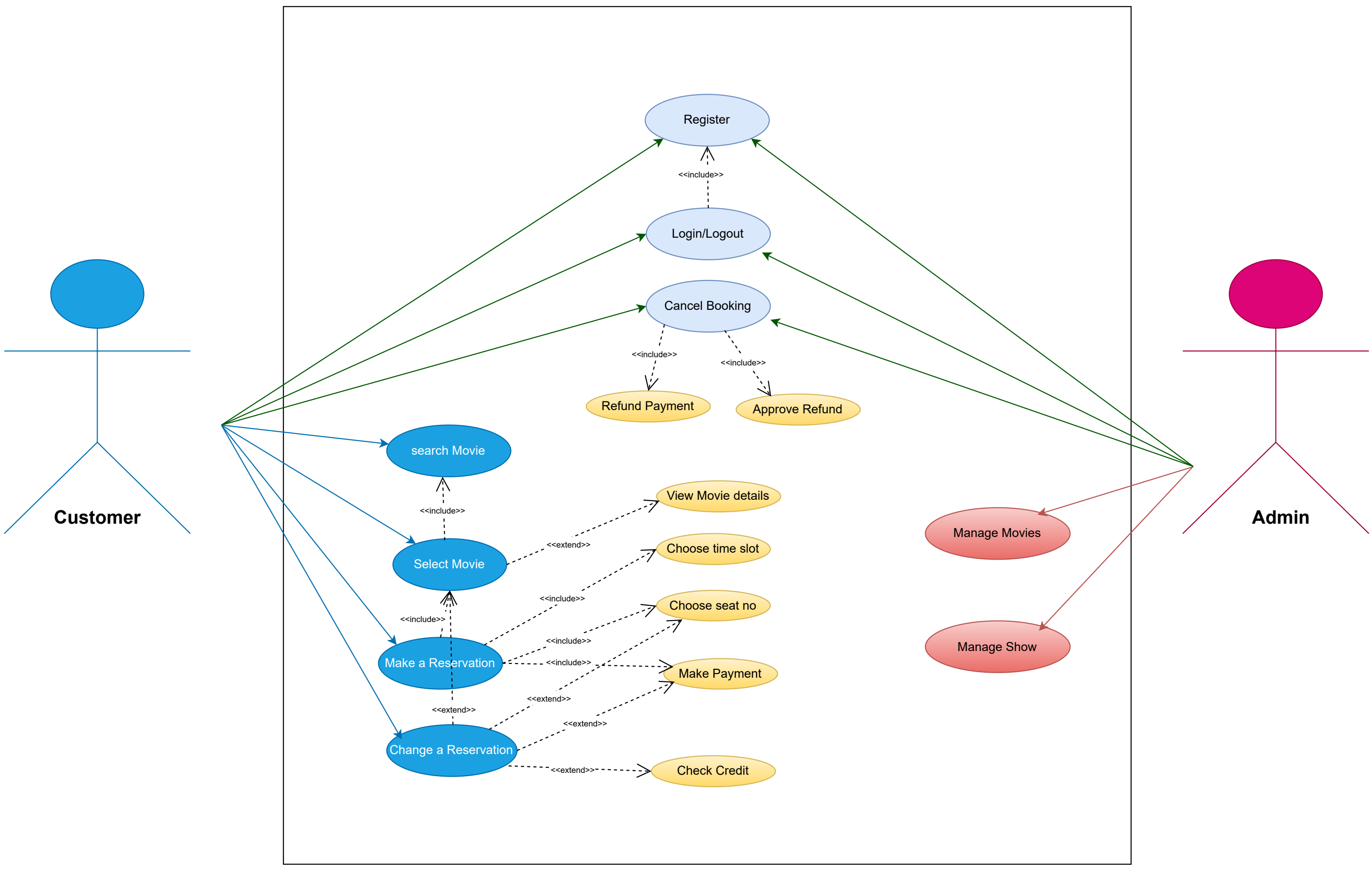
Maintainability (80%):
The system should be designed with a **modular architecture** and well-documented, **clean code** to **facilitate maintenance and updates**. Integration with version control systems and automated testing helps improve long-term maintainability.

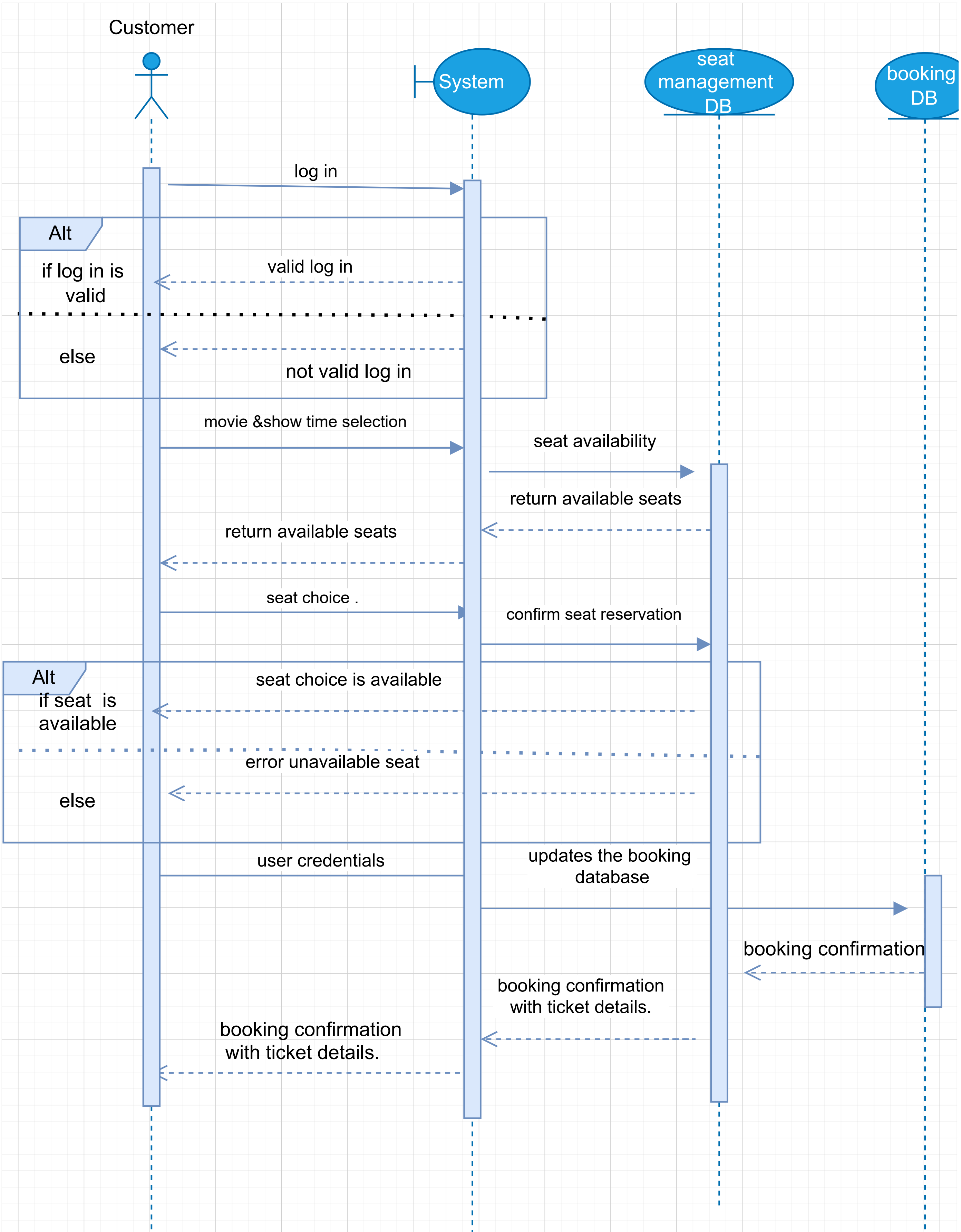
Scalability (85%):

The system should be capable of **handling a growing number of users** and bookings **without performance degradation**. This can be achieved through scalable database solutions, caching techniques, and cloud computing.

Compatibility (70%):

The system should be **compatible with various devices and browsers** to ensure easy access via computers and mobile phones. It should also support integration with payment gateways and third-party services.


Register
<<include>>
Login/Logout
Cancel Booking
<<include>>
<<include>>
Refund Payment
Approve Refund
search Movie
<<include>>
View Movie details
<<extend>>
Select Movie
Choose time slot
<<include>>
<<include>>
Choose seat no
<<include>>
Make a Reservation
<<include>>
Make Payment
<<extend>>
<<extend>>
<<extend>>
Change a Reservation
<<extend>>
Check Credit
Customer
Admin
Manage Movies
Manage Show



Customer
System
seat management DB
booking DB
log in
Alt
if log in is valid
valid log in
else
not valid log in
movie &show time selection
seat availability
return available seats
return available seats
seat choice .
confirm seat reservation
Alt
if seat is available
seat choice is available
error unavailable seat
else
user credentials
updates the booking database
booking confirmation
booking confirmation with ticket details.
booking confirmation with ticket details.